E-Commerce Customer Analytics

Data Analysis & SQL Project Report

Python · SQL · Pandas · Matplotlib

2,000	9	621	31%
Total Customers	Features	Churned	Churn Rate

Submitted: April 2026

Subject: Database Management & Data Analysis

■ Project Overview

Introduction

This project analyses the purchasing and churn behaviour of 2,000 e-commerce customers using Python (Pandas, Matplotlib) and SQL. The goal is to uncover patterns that can help a business understand its customers better, reduce churn, and grow revenue.

Objectives

- Understand the overall customer profile — age, gender, income, and spending habits.
- Identify which customers are most likely to churn (leave).
- Segment customers into groups based on their buying behaviour.
- Answer key business questions using SQL queries on the customer table.
- Visualise findings through charts and diagrams for easy understanding.

Dataset Description

The dataset contains 2,000 customer records with 9 columns covering demographics, purchase behaviour, and churn status. Each row represents one unique customer.

Column	Type	Description
Customer_ID	Integer	Unique identifier for each customer
Age	Integer	Customer age (18–69 years)
Gender	Text	Male or Female
Annual_Income	Integer	Yearly income in currency units
Spending_Score	Integer	Score from 1–99 reflecting spending habits
Membership_Years	Float	Number of years as a member
Online_Purchases	Integer	Total online purchases made (1–199)
Discount_Usage	Float	Fraction of purchases using a discount (0–1)
Churn	Binary	1 = Customer left, 0 = Customer retained

■ Data Summary & Key Statistics

Key Performance Indicators

The table below shows the most important summary statistics computed from the full dataset of 2,000 customers.

Metric	Value	Insight
Total Customers	2,000	Full dataset size
Average Age	43.8 years	Middle-aged customer base
Male / Female Split	956 / 1,044	Slightly more female customers
Avg Annual Income	■84,089	Upper-middle income segment
Avg Spending Score	50.4 / 100	Moderate spenders on average
Avg Membership Years	5.2 years	Mostly long-term members
Avg Online Purchases	98 purchases	High purchase frequency
Churned Customers	621 (31%)	Significant churn risk
Retained Customers	1,379 (69%)	Majority are retained
Highest Spending Score	99 (Cust #114)	Top performer
High Discount + Low Spend	150 customers	Discount abuse risk group

Statistical Distribution

Feature	Min	Max	Mean	Std Dev
Age	18	69	43.8	14.9
Annual Income	20,019	1,49,878	84,089	37,476
Spending Score	1	99	50.4	28.6
Membership Years	0.1	10.0	5.2	2.8
Online Purchases	1	199	98.0	57.3
Discount Usage	0.00	1.00	0.49	0.28

■ Exploratory Data Analysis – Demographics

Age Distribution of Customers

The histogram below shows how customers are spread across different age groups. The largest group (483 customers) is aged 18–30, showing a strong young adult base. All other age groups are fairly balanced between 350 and 425 customers each.

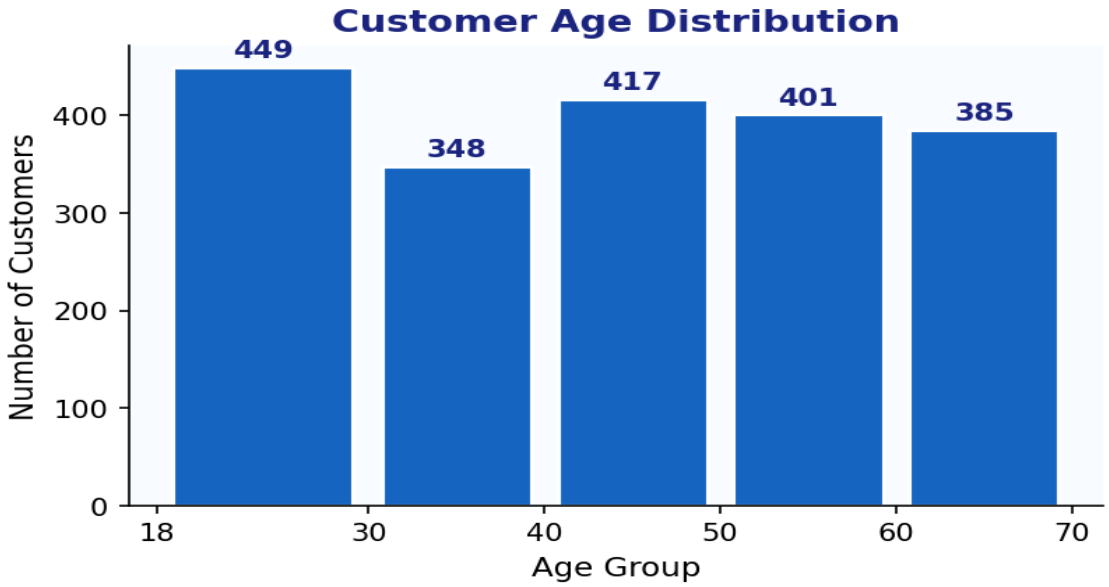

Figure 1 – Customer Age Distribution Histogram

Customer Churn Distribution

31% of customers (621 out of 2,000) have churned, meaning they stopped using the service. This is a significant proportion and warrants targeted retention strategies. 69% of customers remain active.

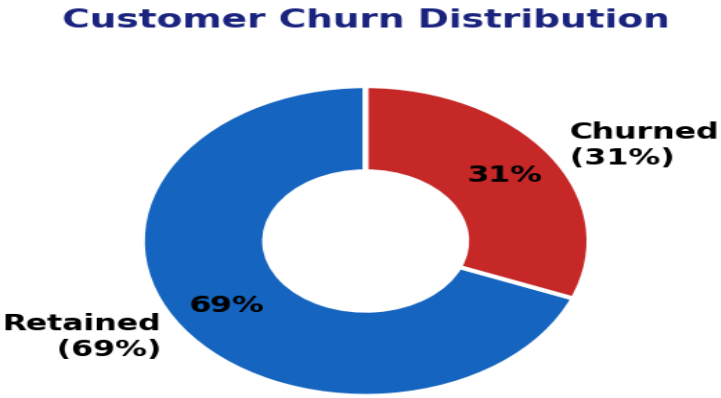

Figure 2 – Churn vs Retained Customers

■ Exploratory Data Analysis – Behaviour

Gender-Wise Comparison

Both genders show very similar average income (■83,795 for Female vs ■84,411 for Male) and spending scores (~50 each). However, Female customers made significantly more total online purchases (103,070) compared to Male customers (92,882).

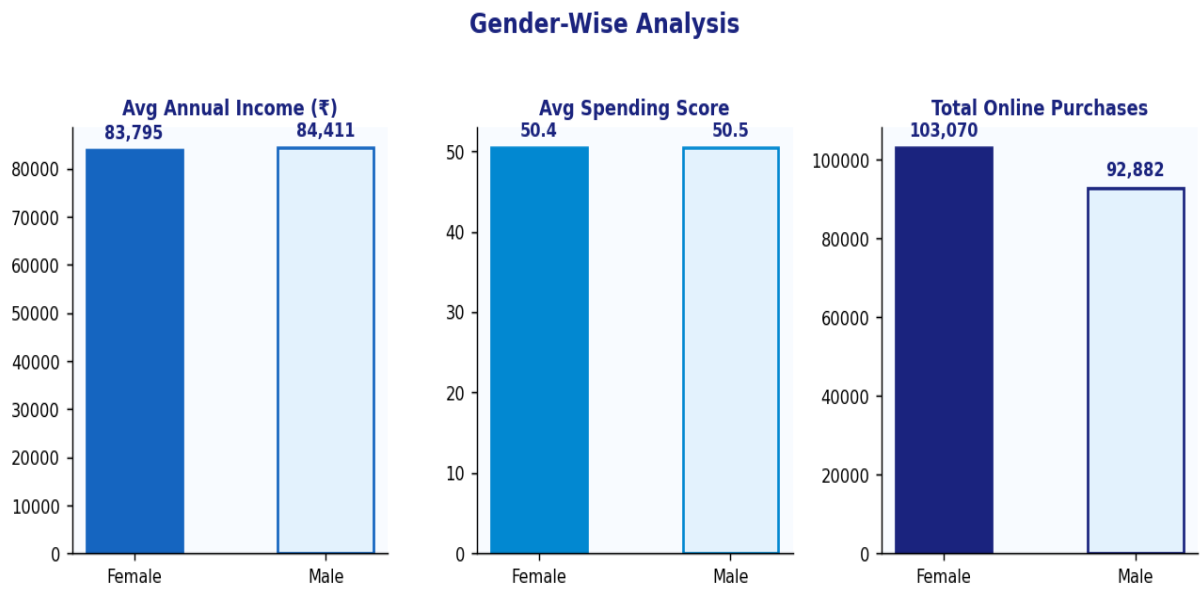

Figure 3 – Gender-Wise: Income, Spending Score & Purchases

Income vs Spending Score (Churn Analysis)

The scatter plot below shows each customer's Annual Income against their Spending Score, colour-coded by churn status. No single income level predicts churn — churned customers (red) are spread across all income brackets, suggesting that factors like discount usage or membership years may be better predictors.

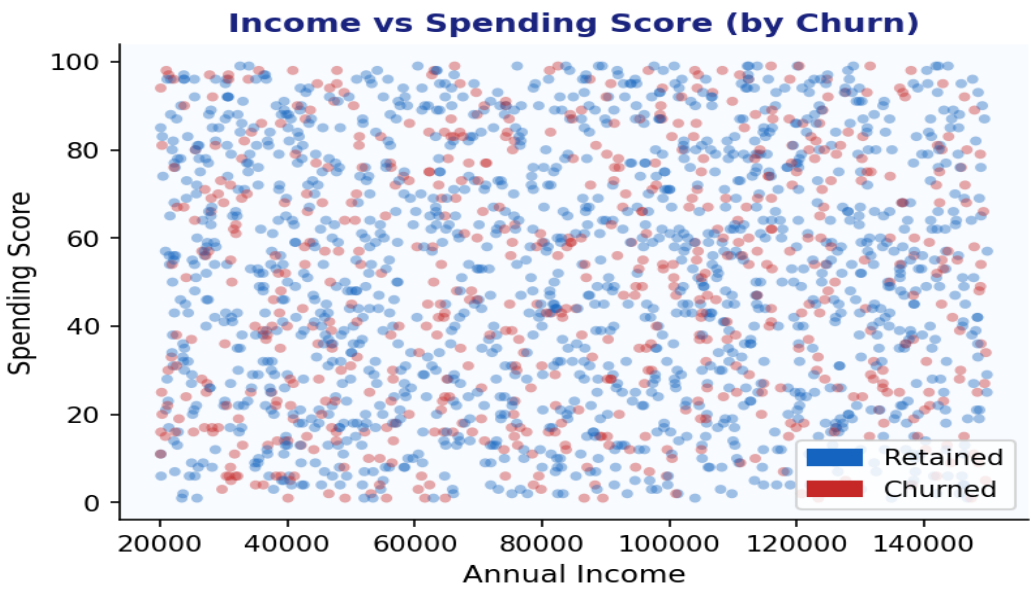

Figure 4 – Income vs Spending Score coloured by Churn

■■ Customer Segmentation

Segmentation Logic

Customers are divided into two segments based on the number of online purchases they have made. This is a simple, data-driven way to identify high-value (Loyal) versus lower-engagement (New) customers.

Segment	Rule	Count	Key Action
■ New	Online Purchases ≤ 150	1,515	Encourage repeat purchases with offers
■ Loyal	Online Purchases > 150	485	Reward with VIP perks & early access

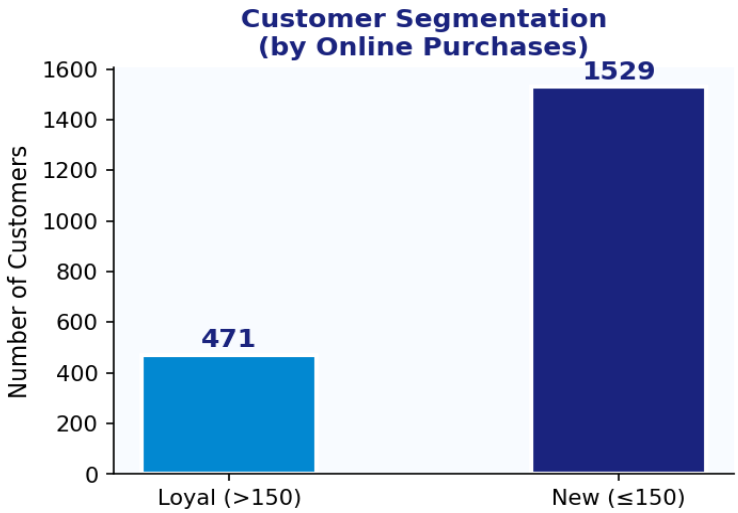

Figure 5 – Customer Segments by Online Purchase Frequency

Key Insights from Segmentation

- ■ 75.8% of customers (1,515) fall in the "New" segment — there is large growth potential.
- ■ 24.2% of customers (485) are already "Loyal" high-frequency buyers.
- ■ Loyal customers likely generate a disproportionately high share of revenue.
- ■ Business should focus on converting "New" customers into "Loyal" ones.
- ■ 150 customers have high discount usage but low spending score — a risk group for discount abuse that should be monitored.

■■ SQL Query Analysis – Part 1

Q1 – Count Total Customers

Simply counts every row to confirm the full dataset size.

```
SELECT COUNT(customer_id) AS total_customers
FROM customer;
```

■ **Result: 2,000 customers**

Q2 – Male Customers Over 50

Filters using two conditions — gender and age. Useful for targeted marketing campaigns for older male users.

```
SELECT * FROM customer
WHERE gender = 'Male' AND age > 50;
```

■ **Result: Returns all male customers aged 51–69**

Q3 – Average Annual Income

The average income helps set appropriate price points and discount thresholds.

```
SELECT AVG(Annual_Income) AS Average_Income
FROM customer;
```

■ **Result: ■84,089.4**

Q4 – Churned Customer Count

Churn = 1 marks a customer who left. This query quantifies the attrition problem.

```
SELECT COUNT(*) AS Churned_Customers
FROM customer WHERE Churn = 1;
```

■ **Result: 621 customers churned (31%)**

Q5 – Customer with Highest Spending Score

Identifies the top spender for VIP recognition or case-study analysis.

```
SELECT customer_id, spending_score
FROM customer
ORDER BY spending_score DESC
LIMIT 1;
```

■ **Result: Customer #114, Score = 99**

■■ SQL Query Analysis – Part 2

Q6 – Gender-Wise Spending & Purchases

Groups data by gender. Reveals that female customers buy more overall.

```
SELECT Gender, AVG(Spending_Score), SUM(Online_Purchases)
FROM customer GROUP BY Gender;
```

■ Female: Avg Score ~50.4, Purchases 103,070 | Male: Avg Score ~50.5, Purchases 92,882

Q7 – Mid-Tenure High-Income Customers

Useful for premium loyalty rewards targeting stable, high-income members.

```
SELECT customer_id, age, gender FROM customer
WHERE membership_years BETWEEN 5 AND 10
AND annual_income > 70000;
```

■ Returns customers with 5–10 years membership and income above ■70,000

Q8 – Top 5 Online Purchasers

Identifies the most active customers for special engagement programs.

```
SELECT customer_id, online_purchases FROM customer
ORDER BY online_purchases DESC LIMIT 5;
```

■ Customers with 199, 198, 197... purchases — the platform power users

Q9 – High Discount, Low Spending

These customers use discounts frequently but spend little. They may be deal-hunters who are unlikely to buy at full price — a segment to watch.

```
SELECT * FROM customer
WHERE discount_usage > 0.7 AND spending_score < 30;
```

■ Result: 150 customers match this pattern

Q10 – Customer Segmentation (New vs Loyal)

Uses a CASE statement to create a derived column. Segments all customers into meaningful business groups without changing the underlying data.

```
SELECT
CASE WHEN online_purchases <= 150 THEN 'NEW'
ELSE 'LOYAL' END AS segment,
COUNT(*) AS count
FROM customer GROUP BY segment;
```

■ **NEW: 1,515 customers | LOYAL: 485 customers**

Conclusion & Business Recommendations

■ **Churn (31%) needs urgent attention.**	Launch re-engagement emails for churned customers and exit-intent pop-ups.
■ **Female customers buy more.**	Design more female-targeted campaigns and product categories.
■ **485 Loyal customers are your goldmine.**	Introduce a VIP loyalty programme with exclusive perks.
■■ **150 discount-abusers risk margin erosion.**	Cap discount frequency per customer or introduce minimum spend rules.
■ **5.2 avg membership years shows stability.**	Leverage tenure in messaging — "Thank you for 5 years with us!"